

Cross-Modal Attention Acts as a Frequency Filter

Why Verbose Prompts Improve Robustness in Vision–Language Models

Anonymous ARR Submission

Qwen3-VL {2B, 8B} & LLaVA-OneVision · GQA & CLEVR · 312k measurements/run

One-sentence summary

Cross-modal attention in a VLM forms a **task-conditioned frequency filter**; its overlap with a corruption's spectrum **predicts drift**, and **verbose prompts broaden the filter to improve robustness**.

Practical recipe: pad the prompt.

Motivation

Vision-language models are fragile under image corruption

A small image perturbation — blur, JPEG, a few-pixel shift — can flip a VLM's answer, even when a human sees no difference.

The puzzle: the *same image* and a *semantically identical* question can be more or less robust depending only on **how the question is worded**.

We ask: *what property of the prompt controls robustness, and why?*



A GQA image under Translation(+4) — our smallest geometric perturbation.

Two opposite effects of question wording

Verbosity $\Rightarrow$ more robust

"Is there a cat?"

$\rightarrow$ *"Please look carefully and answer: is there a cat?"*

Same answer, **much smaller drift**.

Complexity $\Rightarrow$ more fragile

"Is there a cup?"

$\rightarrow$ *"What colour is the cup left of the chair?"*

Finer-grained reasoning, **larger drift**.

Both effects come from **one mechanism** — a frequency filter induced by the prompt.

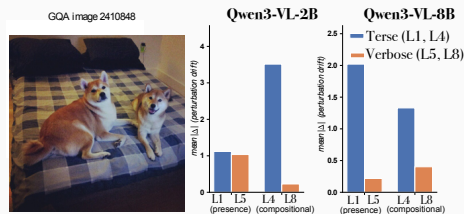
The core idea

Cross-modal attention as a frequency filter

The question's attention over image patches, projected into the 2D Fourier domain and radially binned, is a **distribution over spatial frequencies** — the filter $W_t(\omega)$.

- Low bands = coarse layout / shape
- High bands = fine detail / texture
- Verbose prompts **widen** $W_t(\omega)$
- Fine reasoning **narrows** $W_t(\omega)$

The answer **drifts most** when the filter $W_t(\omega)$ and the corruption's spectrum $\Delta F(\omega)$ sit on the **same frequencies**.



L1 (terse): Is there a bed in this image?

L5 (verbose): Please read the question carefully and answer it directly . Is there a bed in this image. The extra wording is only polite framing.

L4 (terse): What color is the wall that is to the right of the outlet?

L8 (verbose): Please read the question carefully and answer it directly . What color is the wall that is to the right of the outlet? The extra wording is only polite framing.

Verbose paraphrases cut drift variance across corruptions.

The mechanism in one picture

prompt $\longrightarrow$ cross-modal attention $\longrightarrow$ filter $W_t(\omega)$ over visual frequencies

corruption $\longrightarrow$ spectral signature $\Delta F(\omega)$

drift $\propto \langle W_t(\omega), \Delta F(\omega) \rangle$ (spectral overlap)

- $W_t(\omega)$ is read from attention — *no behavioural fitting*.
- $\Delta F(\omega)$ is a model-free radial power spectrum of the pixel difference.
- The overlap is a **parameter-free** predictor of robustness.

Design

The mirror 8-level design

Eight questions per image, separating **semantic complexity** (C_{sem}) from **prompt length** (C_{prompt}) on orthogonal axes.

L1–L4: terse, increasing granularity

- L1: object presence (Y/N)
- L2: attribute query (MCQ)
- L3: spatial relation (Y/N)
- L4: compositional (MCQ)

L5–L8: verbose mirrors

- Same semantic atoms, byte-for-byte
- Inflate C_{prompt} *alone*
- Polite framing wrapper

Grid: $\{\text{Qwen3-VL-2B}, \text{Qwen3-VL-8B}\} \times \{\text{GQA}, \text{CLEVR}\}$, $n = 500/\text{cell}$, severity $\{1,2,3\}$, 78 perturbation slices/image = 312k scoring rows/run.

Four predictions

Four separable predictions — all confirmed

- P1 Tug-of-war:** complexity $\uparrow$ destabilises, verbosity $\downarrow$ stabilises — opposite signs at the item level.
- P2 Bandwidth trajectory:** $W_t(\omega)$ narrows then broadens across decoder layers; verbose mirrors end up broader.
- P3 Overlap law:** $\langle W_t(\omega), \Delta F(\omega) \rangle$ predicts drift, positive in **12/12** cells.
- P4 Cutoff reframed:** most corruptions are low-frequency; $W_t(\omega)$ puts only 3–5% mass on the top band.

Tested on different observables $\Rightarrow$ they don't share noise. The central claim survives setting any one aside.

P1 — The tug-of-war on volatility

Within-image fixed-effects regression (Frisch–Waugh–Lovell), z-scored:

$$V_{i,t} = \beta_{C_{\text{res}}} C_{\text{res}}(t) + \beta_{C_{\text{prompt}}} C_{\text{prompt}}(t) + \beta_H H_{\text{opt}}(t) + \alpha_i + \varepsilon_{i,t}$$

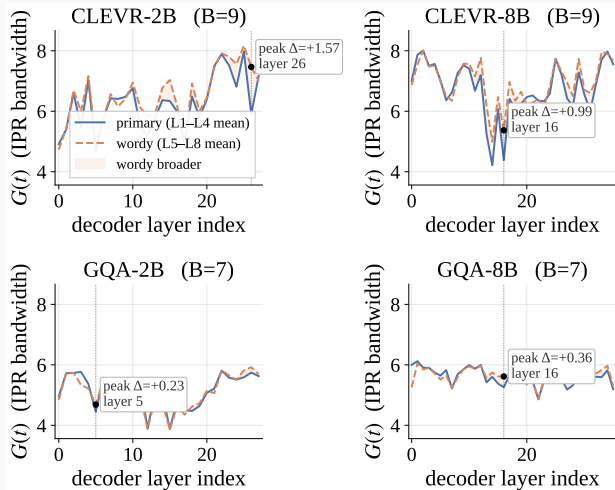
Model	Bench	$\tilde{\beta}_{C_{\text{res}}}$	$\tilde{\beta}_{C_{\text{prompt}}}$
Qwen-2B	CLEVR	+0.447	−0.330
Qwen-8B	CLEVR	+0.257	−0.749
Qwen-2B	GQA	+0.290	−0.289
Qwen-8B	GQA	+0.075	−0.623

✓ $\tilde{\beta}_{C_{\text{res}}} > 0$ in 4/4 (complexity destabilises)

✓ $\tilde{\beta}_{C_{\text{prompt}}} < 0$ in 4/4 (verbosity stabilises)

CR1 on 312k rows: $\beta_{C_{\text{sem}}} > 0$ at $p \leq 10^{-6}$,
 $\beta_{C_{\text{prompt}}} < 0$ at $p \leq 10^{-64}$.

P2 — Bandwidth narrows then broadens across depth

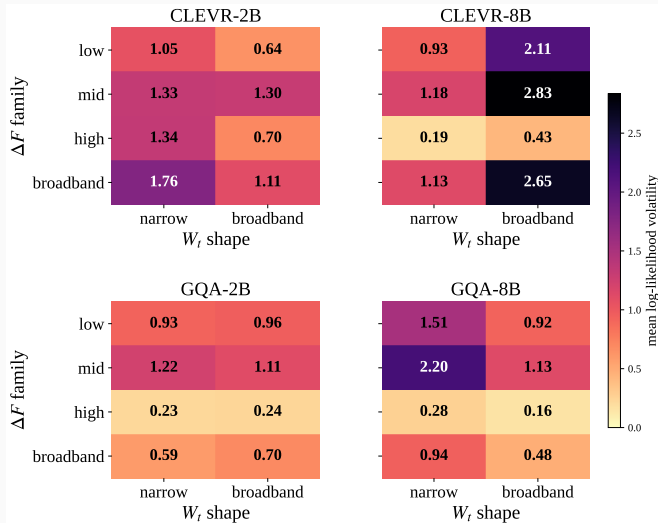


Four-stage trajectory in all runs:

1. **Layer 0:** length prior (verbose < terse)
2. **Early:** prior erasure
3. **Mid:** C_{sem} -driven narrowing
4. **Late:** verbosity **widening**

Run	$G(t)$ at last_2	
	terse	verbose
CLEVR-2B	5.90	7.47
GQA-8B	5.82	5.97

P2 — Narrow vs. broadband filters drift differently

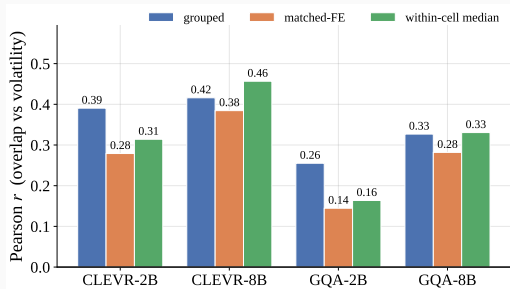


Narrow- $W_t(\omega)$ tasks lose more on a band-matched perturbation than broadband- $W_t(\omega)$ tasks.

✓ Narrow > broadband in **4/4** rows on CLEVR-2B and GQA-8B.

CLEVR-8B reverses — the recovery-dominance regime where verbose mirrors sit at the log-prob ceiling.

P3 — The overlap law predicts drift

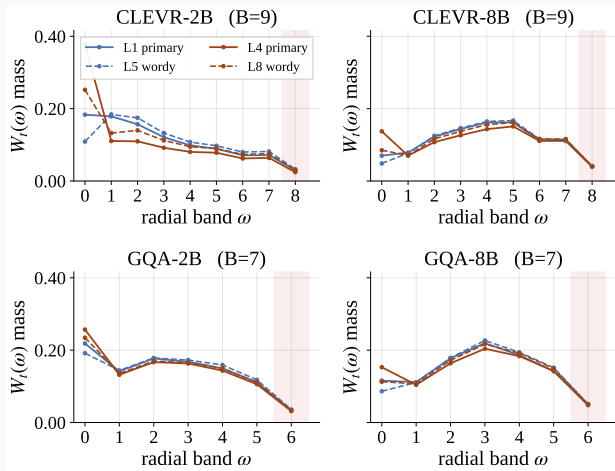


Model	Bench	grp	FE	cell
Qwen-2B	CLEVR	.39	.28	.31
Qwen-8B	CLEVR	.42	.38	.46
Qwen-2B	GQA	.26	.15	.16
Qwen-8B	GQA	.33	.28	.33

✓ **Positive in 12/12** (run $\times$ aggregation)
cells, $p \approx 0$ everywhere.

Parameter-free: no behavioural calibration.
Grouped r ranges 0.26–0.42.

P4 — High-frequency noise is not the hard case



Common assumption: VLMs are most vulnerable to high-frequency noise.

We find the opposite:

- Most real perturbations are **low-frequency**.
- $W_t(\omega)$ puts only **3–5%** mass on the top band.
- $\Rightarrow$ high-freq corruptions cause the **smallest** drift in 3/4 runs.

What older work calls “high” is our mid-band.

Cross-architecture

The framework is architectural, not Qwen-specific

Same pipeline on **LLaVA-OneVision-0.5B** (SigLIP + Qwen2, no CLS, $B = 20$), GQA, 500 images.

	Qwen-2B (GQA)	Qwen-8B (GQA)	LLaVA-OV-0.5B (GQA)
Overlap r (last_2)	+0.255	+0.327	+0.343
last_2 $G(t)$ terse/verbose	5.75/5.92	5.82/5.97	15.72/ 16.28
$\tilde{\beta}_{C_{\text{prompt}}}$	-0.289	-0.623	-0.140
Top-band mass	3-5%	3-5%	< 1%

All four predictions replicate on a different VLM family — the trajectory, overlap law, cutoff, and the verbose/complexity signs.

Takeaways

What this buys you

- **A mechanism, not just a benchmark:** robustness is governed by spectral alignment between a prompt-induced filter and the corruption.
- **A parameter-free predictor:** $\langle W_t(\omega), \Delta F(\omega) \rangle$ forecasts drift from model internals — no held-out calibration.
- **A reframed cutoff law:** low-frequency, not high-frequency, corruptions dominate VLM fragility.
- **A free robustness lever:** verbose paraphrasing reduces drift variance by **70–81%** on the 8B model.

Pad the prompt.

Limitations & honesty

- Results on Qwen3-VL- $\{2B, 8B\}$; LLaVA-OneVision is a **single-condition** cross-architecture check (0.5B, GQA only).
- Some gates hold cleanly only on 8B (signed-drift reverses on CLEVR-8B under recovery dominance).
- Overlap law stronger on CLEVR; cutoff law cleaner on GQA.
- $W_t(\omega)$ is read from attention scores — a **correlational** lower bound on the causal filter.
- Verbose mirrors are programmatic; human paraphrases are future work.

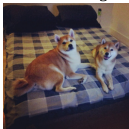
Cross-modal attention is a frequency filter.

Verbose prompts broaden it; fine reasoning narrows it;
overlap with the corruption controls drift.

Pad the prompt.

Backup — the pipeline, end to end

Source image



Scene-graph snippet (input to the generator)

objects: {dog (x2), bed, wall, pillow, outlet, sheet, ...}

attributes: wall.color = white; pillow.color = blue; bed = large

relations: dog on bed; pillow on bed; wall to-the-right-of outlet

L1/L3 alternate positive and negative phrasings by image index; L4 reuses L2's options unchanged (v8 design).

L1: presence

"Is there a bed in this image?"

gold: (A) yes

L2: attribute

"What color is the wall?"

gold: (C) white
options: {white, +3 distractors}

L3: relation (neg.)

"Is the outlet to the right of the wall?"

gold: (B) no

L4: compositional

"What color is the wall that is to the right of the outlet?"

gold: (C) white
options inherited from L2

L5: wordy of L1

"Please read the question carefully and answer it directly. primary question The extra wording is only polite framing."

same options, same gold

L6: wordy of L2

"Please read the question carefully and answer it directly. primary question The extra wording is only polite framing."

same options, same gold

L7: wordy of L3

"Please read the question carefully and answer it directly. primary question The extra wording is only polite framing."

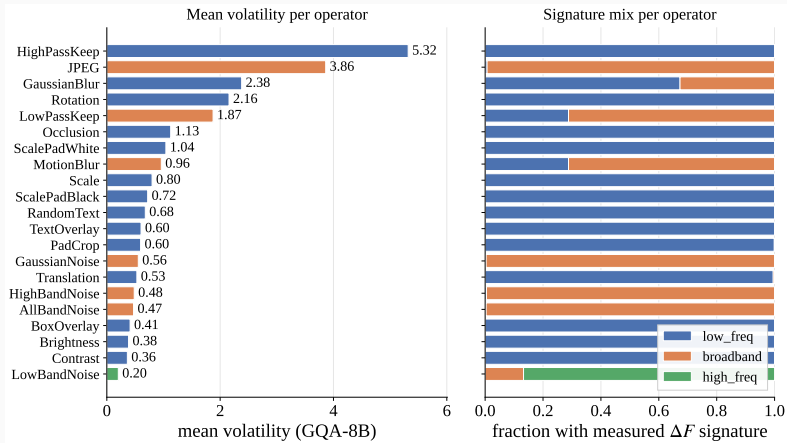
same options, same gold

L8: wordy of L4

"Please read the question carefully and answer it directly. primary question The extra wording is only polite framing."

same options, same gold

Backup — designed vs. measured spectral signatures



All analyses condition on the *measured* $\Delta F(\omega)$, not the operator's nominal name.

Backup — worked example (one GQA image)

Clean



Translation(+4)



Verbose mirrors arrive near 0 nat; terse primaries up to 19 nat lower — motivating within-cell volatility as the analysis target.

Future directions

Direction 1 — Spectral attention control for global-configuration tasks

Two documented clean-image failures of VLMs:

- **Visual path following:** models lose the target path and switch to *locally similar* distractors — “local competition” pulls them off the true continuation (Hong et al., 2026; arXiv:2605.15672).
- **Analog clock reading:** models confuse hour/minute hands and miss the *global hand configuration* (ClockBench: best model 13.3% vs. humans 89.1%) (*It's Time to Get It Right*, arXiv:2603.08011; *Lost in Time*, arXiv:2502.05092).

Hypothesis. Both failures share a cause: attention **over-localises** on salient but incomplete local features, losing global relational context. Our work shows **narrow** $W_t(\omega) \rightarrow \text{fragile}$ and **broad** $W_t(\omega) \rightarrow \text{robust \& higher clean confidence}$ — suggesting a controllable lever.

Direction 1 — Proposed test & the honest tension

Proposed intervention

Spectrally **broaden** the attention filter (shift mass toward low bands = global layout) at decode time, and measure tracing / clock-reading accuracy on clean images.

The open question (not assumed away)

In our data, **fine-grained reasoning narrows** $W_t(\omega)$ — it *needs* local detail. So broadening universally risks hurting local discriminability.

The real test: can we add **global awareness without losing local discriminability**? That trade-off *is* the tug-of-war (C_{sem} narrows vs. C_{prompt} broadens) measured directly on a task where global structure is the bottleneck.

Direction 2 — Vision's depth advantage vs. its token disadvantage

Established (width disadvantage):

- Vision tokens are low information-density; VLMs need *many* of them.
- Attention to vision tokens *drops* during text generation (e.g. arXiv:2411.03312).

Underexplored (depth):

- Visually-grounded predictions **stabilise by middle layers** — $\sim$ layer 16 (POPE, Y/N), $\sim$ 20 (GQA) of $\sim$ 29.
- Visual representations become *interchangeable* across deeper layers once stabilised (logit-lens / early-exit studies).

Reframing hypothesis. Vision's **width** disadvantage (more tokens) may be partly offset by a **depth** advantage — fewer *layers* needed for a visually-grounded answer $\Rightarrow$ early-exit compute savings.

Link to our P2: the attention filter $W_t(\omega)$ *stabilises its shape by mid/late layers* — visual processing largely completes before the final layers, consistent with early convergence.

Direction 2 — Proposed measurements

1. **Per-layer answer stability:** logit-lens the gold answer at every layer; find the depth ℓ^* where the prediction stops changing, separately for *vision-grounded* vs. *text-only* items.
2. **Filter convergence depth:** the layer where $W_t(\omega)$'s shape (bandwidth $G(t)$, band-mass) stops moving — does it coincide with ℓ^* ?
3. **Width–depth trade-off:** under a fixed compute budget, trade vision-token count (width) against layers used (depth); map the Pareto frontier.

If visually-grounded answers converge early, **early exit on vision-heavy items** recovers compute that the token disadvantage spends — a depth-side answer to a width-side problem.

Two levers from one mechanism:

broaden the filter for global-configuration tasks,
exit early where the filter has already converged.

Both testable with the tools in this paper.